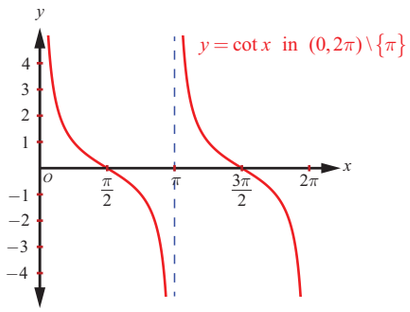


Fig. 4.27

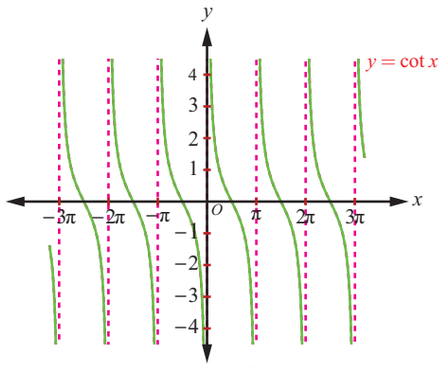


Fig. 4.28

The graph of $y = \cot x$, $x \in (0, 2\pi) \setminus \{\pi\}$ is shown in Fig 4.27. The same segment of the graph of cotangent for $(0, 2\pi) \setminus \{\pi\}$ is repeated for $(2\pi, 4\pi) \setminus \{3\pi\}, (4\pi, 6\pi) \setminus \{5\pi\}, \cdots$, and for $\cdots$, $(-4\pi, -2\pi) \setminus \{-3\pi\}$, $(-2\pi, 0) \setminus \{-\pi\}$. The entire graph of cotangent function with domain $\mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$ is shown in Fig. 4.28.

4.8.2 Inverse cotangent function

The cotangent function is not one-to-one in its entire domain $\mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$. However, $\cot : (0, \pi) \to (-\infty, \infty)$ is bijective with the restricted domain $(0, \pi)$. So, we can define the inverse cotangent function with $(-\infty, \infty)$ as its domain and $(0, \pi)$ as its range.

Definition 4.8

The inverse cotangent function $\cot^{-1} : (-\infty, \infty) \to (0, \pi)$ is defined by $\cot^{-1}(x) = y$ if and only if $\cot y = x$ and $y \in (0, \pi)$.

4.8.3 Graph of the inverse cotangent function

The inverse cotangent function, $y = \cot^{-1} x$ is a function whose domain is $\mathbb{R}$ and the range is $(0, \pi)$. That is, $\cot^{-1} x : (-\infty, \infty) \to (0, \pi)$.

Fig. 4.29 and Fig. 4.30 show the cotangent function in the principal domain and the inverse cotangent function in the corresponding domain, respectively.

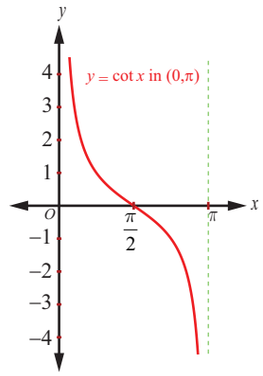


Fig. 4.29

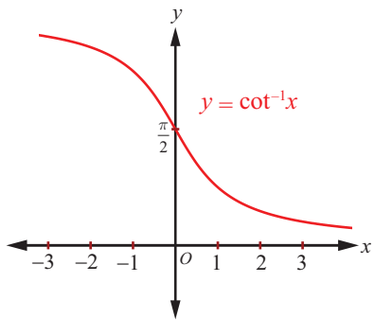


Fig. 4.30

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